

# Aviral Sharma

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## Summary

Machine learning engineer specializing in building and deploying AI-driven product authentication systems. Experienced in cloud-based MLOps pipelines, computer vision, deep learning, data engineering workflows, real-time analytics, and scalable production deployment using tools like AWS, TensorFlow, MLflow, and Docker.

## Experience

### ML Engineer

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Dec 2024 – Present

- Developed and deployed computer vision models for QR code-based product authentication using Vision Transformers (ViT), contrastive learning, and transfer learning on datasets exceeding 200K images.
- Built automated training and evaluation pipelines on AWS SageMaker, leveraging distributed training, hyperparameter tuning, and experiment tracking with MLflow.
- Designed and deployed real-time inference endpoints on AWS, achieving 97.9% authentication accuracy and reducing average prediction latency by 65% through optimized model compression and batch processing.
- Implemented scalable MLOps workflows using Airflow (Astronomer), automating data ingestion, preprocessing, model retraining, validation gating, and deployment workflows.
- Created observability and monitoring dashboards using Grafana and AWS CloudWatch to track model drift, pipeline performance, and system reliability.
- Containerized complete ML services using Docker and orchestrated production deployments on Kubernetes clusters to ensure high availability and fault tolerance.
- Collaborated with backend and devops teams to design secure inference APIs, CI/CD pipelines, and model artifact management using Docker, ECR, and S3.

## Key Projects

### QR Code Authentication Model

- Designed a Vision Transformer-based (ViT) model for Copy Detection Pattern (CDP) recognition and forgery detection.
- Deployed the model on AWS SageMaker with automated training, validation pipelines, and real-time inference endpoints.
- Technologies:** Python, PyTorch, TensorFlow, ViT, AWS SageMaker, MLflow, Docker

### End-to-End ML Deployment Pipeline

- Built a complete MLOps workflow covering dataset ingestion, preprocessing, model training, experiment tracking, CI/CD, and production deployment.
- Implemented automated retraining with Airflow, integrated drift detection alerts, and set up scalable inference endpoints on AWS EKS (Kubernetes).
- Developed monitoring dashboards to track latency, throughput, model drift, and system health using Grafana and CloudWatch.
- Technologies:** AWS EKS, Docker, Kubernetes, Airflow, MLflow, CloudWatch, Grafana, REST APIs

## Achievements

- Achieved **97.9% accuracy** and cut inference latency by **65%** through ViT-based model optimization on AWS.
- Built an automated **end-to-end MLOps pipeline** with SageMaker, Airflow, and MLflow for training, tuning, and retraining.
- Deployed scalable ML services using **Docker & Kubernetes (EKS)** with monitoring via Grafana and CloudWatch.
- Improved infrastructure reliability by integrating secure inference APIs, CI/CD workflows, and model versioning with **Docker, ECR, and S3**.

## Technologies

- Machine Learning:** Python, PyTorch, TensorFlow, Vision Transformers, Scikit-learn, OpenCV
- MLOps & Cloud:** AWS SageMaker, AWS Lambda, AWS S3, AWS CloudWatch, MLflow, Airflow, Docker, Kubernetes, Grafana
- DevOps & Tools:** REST APIs, CI/CD, GitHub Actions

## Education

Dr. A. P. J. Abdul Kalam Technical University

Sept 2020 – July 2024

Bachelor of Technology in Information Technology

## Publications

Deep Learning-Based Accurate and Efficient Human Tracking and Identification

July 2025

Kaushal Kishor, *Aviral Sharma*, Aryash

doi.org/10.1007/978-981-96-3102-5\_53